

DHARMIN SHAH

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SUMMARY

Engineering leader and hands-on architect with 10+ years building large-scale backend and data platforms in Java, Scala, and Python — a decade of it in security engineering across threat intelligence, email security, and AI safety. Most recently architected Microsoft's AI safety data platform ingesting 1B+ records per day from 60+ AI services, and shipped LLM-based agents and security tooling that detect and measure AI-specific threats, including direct and indirect prompt injection and jailbreaks. Equally effective as a principal engineer or a manager growing teams of 8+; known for turning ambiguous safety and security problems into shipped, measurable systems.

EXPERIENCE

Staff Software Engineer | Ent.ai

July 2026 – Present

- Staff software Engineer on the platform team

Principal Software Engineer | Microsoft — AI Safety Platform

April 2025 – July 2026

- Architected a greenfield AI safety data platform (Azure Databricks, Data Factory, Data Explorer) ingesting **1B+ records/day** of threat telemetry from **60+ AI services**, delivering a unified cross-model view of AI threats and tracking **20+ safety metrics** — including attack rates for direct and indirect prompt injection and jailbreaks.
- Built a natural-language threat-hunting agent (Python, Java) that converts analyst queries to KQL and autonomously executes follow-up queries via RAG — cutting investigation time **from weeks to hours** and enabling Microsoft's Digital Crimes Unit to continuously monitor threat-actor infrastructure so investigators could focus on the hunt itself.
- Built an MCP server for security stress-testing of AI systems; partnered with safety researchers and product teams to translate ambiguous safety concerns into automated, evaluable detection signals.
- Defined product SLOs/SLAs and built observability dashboards and incident-response workflows for Tier-0 safety pipelines; advised partner teams on building their AI infrastructure and products securely.
- Contributed to *Clarity Agent*, a Microsoft open-source AI thinking partner that helps users reason through projects rather than jumping straight to answers.

Engineering Manager | Microsoft — Unified Security Intelligence

March 2022 – April 2025

- Managed a distributed team of **8 engineers** building data collection systems across petabyte-scale intelligence stores; grew the team by promoting multiple engineers from junior to senior and one senior engineer to staff.
- Led re-architecture of global backend data pipelines (Java, Spark, Kafka) processing TB-scale threat intelligence, improving data freshness from **80% coverage in 60 days to 95% in 14 days**.
- Cut production incident (IcM) tickets **from several hundreds to sub-hundreds** by shipping self-healing systems and better error handling — transforming on-call from an SME-dependent burden into a rotation any engineer could triage independently.
- Drove datacenter-to-cloud migration and established SLA tracking, alerting, and centralized observability, reducing data refresh **from 100+ days to 14 days** for 95% of domains.
- Defined technical vision for region-agnostic deployments and automated production operations, eliminating hundreds of manual engineering touches per quarter.

Engineering Manager | Proofpoint Inc.

July 2021 – March 2022

- Led a four-person team improving API performance, user experience, and service uptime for enterprise email security products.
- Set the product roadmap, balanced feature development against technical-debt reduction, and drove clear next steps across cross-functional stakeholders in an ambiguous environment.

Senior Software Engineer | Proofpoint Inc.

June 2017 – June 2021

- **Nexus Threat Graph (NTX)**: built a multi-TB distributed security data platform from scratch (Java, HBase/Phoenix, Cassandra, Redis) — data models, REST APIs, CI/CD, and infrastructure. Authored the testing framework (**0% → 30% coverage, 70% faster execution**) and drove Python threat clustering to production.

- **CLEAR (email threat analysis & response):** designed Scala APIs orchestrating cross-system email scanning via AWS SQS/S3, with PostgreSQL/Aurora analytics powering customer dashboards; partnered with Data Science on Spark pipelines processing **10 TB**.
- Co-developed Terraform standards and AWS deployment tooling (S3, SQS, DynamoDB, Aurora) adopted across engineering teams.

Software Engineer | Proofpoint Inc.

June 2015 – June 2017

- On a two-person team, built a sales-enablement tool demonstrating product efficacy — a Scala API backed by Cassandra to query threat intelligence for any URL or email attachment; later merged into NTX, where it remains in production with minimal maintenance.
- Built the UI and API (Scala / Play framework) delivering network threat intelligence to enterprise customers.

SKILLS & TECHNOLOGIES

Languages & Frameworks: Python, Java, Scala, SQL; REST APIs, GraphQL, Spring Boot, Node.js

AI / Security: LLM-based systems & agents, RAG pipelines, MCP servers, prompt engineering, Azure OpenAI; AI security tooling, prompt-injection/jailbreak detection, AI red-teaming & stress-testing

Cloud & Infrastructure: AWS (SQS, S3, DynamoDB, Aurora), Azure (Databricks, Data Factory, Data Explorer, DevOps), GCP; Kubernetes, Docker, Terraform, Bicep

Data & Distributed Systems: Spark, Kafka, Hadoop, HBase/Phoenix, Cassandra, PostgreSQL, Redis, NoSQL

Reliability & Ops: SLA/SLO design, observability, on-call & incident response, CI/CD (Azure DevOps, Jenkins); threat intelligence, email security

TALKS & OPEN SOURCE

- Speaker, “*AI Safety and Security*,” Rose-Hulman Institute of Technology.
- *Clarity Agent* — Microsoft open-source project (AI thinking partner).

PERSONAL PROJECTS

- **Personal Finance Dashboard** — Streamlit + PostgreSQL + LLM agents for automated financial insights and natural-language querying.
- **Fishbone Generator** — AI-powered threat-modeling tool using a fishbone-diagram approach to analyze security risks.
- **Real Estate Scouting Agent** — Autonomous agent that browses Redfin listings, applies user preferences, performs image analysis for maintenance items, and generates detailed reports with cost estimates.
- **Cost of Moving Calculator** — Models tax changes and cost-of-living differences to calculate the true cost of relocating within the US.
- **Oura Dash** – Dashboard for tracking your Oura ring metrics. Inspired by cracked-oura repo.

EDUCATION

M.B.A. — Indiana University, Kelley School of Business

B.S., Computer Science & Software Engineering — Rose-Hulman Institute of Technology · *Summa Cum Laude*